

Preliminary Chapter

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These notes are an extract from EYNTKA Algebra ([1]) that represent a detailed “preliminary chapter”. If naïve set theory is new to you, this section is going to go through a lot of terminology. The reader may want to skip to certain sections depending on their background:

1. Sets: sets, power-sets, unions and intersections. Logical quantifiers, basic propositions, vacuous truth. Induction. Relations, equivalence Relations, partial and total orders.
2. Functions: injective/surjective/bijective functions and their relation to equivalence relations. Image and restriction, composition of functions, associativity of composition, Cardinality
3. Number theory: GCD and LCM, divisibility, prime factorization, coprime numbers, the euclidean algorithm, Bézout’s Identity, Modular Arithmetics, Fermat’s Little Theorem
4. Graphs: basic properties
5. Category Theory: intuition, universal property, functors.

You can look over this section if you want a refresher, or skip to section 0.1.4 to start from equivalence relations, in particular if the reader is unaware of the relation between equivalence relations and functions, then this section will be important.

0.1 Sets

As in any field of study, we need to establish a common language for everyone in the field to communicate their ideas. The accepted language of mathematics today was invented by Georg Cantor in the 19th century and is called *set theory*. Before Cantor, there’s wasn’t one universal set of symbols and concepts mathematicians agreed upon to talk about basic concepts, with different groups of mathematician inventing their own symbology to talk about their ideas. Nowadays, every field of mathematics grounds their work in set theory ¹ – algebra is no exception.

We’ll be learning a small part of set theory which concerns itself with using the tools of set theory for talking about concepts. These tools are under the label of *naïve set theory*. We will be using these tools and question little why we are using these particular tools. The reason for the term “naïve” set theory is due to history: there were a couple of important paradoxes which showed some inconsistencies in set theory ². From the existence of these paradoxes, a generation of mathematician’s tried to free set theory from these problems ³. Their research became known as set theory, and the previous understanding of set theory came to be known as *naïve set theory*. Naïve set theory wasn’t dismissed, instead mathematicians who were interested in making a solid foundation for mathematics tried figuring out exactly how we should be using naïve set theory without contradiction, while the rest of mathematicians used naïve set theory. When we use naïve set theory, we’re essentially using the results of [the new] set theory without worrying about the details. It’s like taking advantage of a building without knowing the architectural details of why a support was put here, or a door was

¹There is a movement to replace foundation of mathematics with *type theory* – for those interested see ref:HERE

²Famously, Russel’s paradox showed a logical problem in how we can define sets. We’ll actually cover Russel’s Paradox in ref:HERE (Perhaps add it to the end of set theory section?) if you’re curious to check it out after reading this section

³one major result of this research was the axiomatization of set theory (i.e. the development of the fundamental rules) by Zermelo and Frankel, their set of axioms (i.e. fundamental rules) being called ZF in honour of them. There are a couple of axioms that can be added or removed from ZF, but ZF generally forms the core of modern mathematics. If you keep doing mathematics, you will hear statements like “This statement is true in ZF”

needed there. We will be taking full advantage of this pre-built building for us. Some footnotes will be added for those who want to peek at the questions that lead mathematicians of the early 20th century to investigate the details of set theory. For those with a deeper interest in the foundations of mathematics, a class on logic is recommended, or you can read one of these books: ⁴

As a consequence of using naïve set theory, we'll be defining certain concepts by relying on some [very] basic intuition. For example, we will assume the idea that we can conceptualize a collection of objects (for example, imagine a bunch of trees, or a bunch of numbers), or know the basics of numbers (ex. you know how to think of the symbols 1, 2, 3, ... as number, and you can add/subtract/-multiply/divide). Definition's after the preliminary chapter will build on the definition's presented here, and so it's not a prerequisite for chapters after the preliminary chapter to rely heavily on intuition.

Basic Definitions: Sets

In mathematics, we work with collection of things, and with “mathematical sentences” about a collection of things, where a sentence that must either be true or false. We will start by learning how to describe the collection of things, and then move on to learning about mathematical sentences. We will make a distinction between a collection of things and the things inside the collection. We call a collection of things a *set*, and the things inside a set are *elements*. Another definition of a set is provided by the founder of set theory Georg Cantor

A set is a Many that allows itself to be thought of as a One

In other words, we can think of any arbitrary collection of objects put together as a new single entity. ⁵ The one restriction we give to sets is that all the elements in a set must be different, in other words, we must have each element to be distinct.

When we write a set, we put curly braces (“{” and “}”) around a collection of things (or really, symbols which could represent things ⁶) which are separated by a comma.

Example 0.1.1: Example of sets

1. To write a set with the numbers 1, 2 and 3, we write $\{1, 2, 3\}$. The numbers 1, 2, and 3 are the elements of $\{1, 2, 3\}$. Note that $\{1, 1, 2, 3\}$ would not be considered a set since the 1's repeat. The convention when a set has repeated elements is to have them “squished” into one element to create a set with only distinct elements. This would mean that we would consider the set $\{1, 1\}$ to be $\{1\}$ where we squished the two 1's into a single element. This convention also holds if the set has other elements too:

$$\{1, 1, 2, 3\} = \{1, 2, 3\}$$

If we add an element to a set that's already there, we will also do this “squishing” ^a

⁴here, some good one's in my Book folder

⁵Going deeper than this, asking questions like what are the possible things a set can contain, what are all possible sets, can there be sets that contain itself, all venture into set theory proper

⁶Some might be dubious on whether the symbols can represent “anything”. For example, is it okay to say that x represents a statement that is both true and false? Is that “a thing”. This type of question of what is a “well-defined” thing is subject to deeper explorations of set theory

2. As mentioned, sets contain can anything. The set

$$\{\text{ice-cream, cheesecake, chocolate}\}$$

is a set that represents 3 desserts. The set represents the weekdays of the week in 5 different languages.

3. The set $\{1\}$ is a set with the element 1. We would not say that 1 and $\{1\}$ are the same thing, since $\{1\}$ is “a set containing the number one as its element” and 1 as “the number one”. Using mathematical notation, $1 \neq \{1\}$
4. $\{\}$ is also a set, called the empty set. It pops up often enough that we have a special symbol for this set: $\emptyset$
5. Sets can contain sets, so $\{1, \{\odot, \mathfrak{T}\}, \text{bat}\}$ is a set that contains the element 1, the element $\{\odot, \mathfrak{T}\}$ (which is also a set), and the element bat.
6. Sets need not have an order, so for example

$$\{\text{ice-cream, cheesecake, chocolate}\} = \{\text{cheesecake, ice-cream, chocolate}\}$$

This also means that a set with days of the week (written in any language) is the same if we re-arrange the days:

^aFormally, (explain what’s actually happening here in terms of ZFC?)

Sometimes, we care about the order of elements. If that is the case, we will surround them in parenthesis, “(and)”, and call then n -tuples, where n represents the number of elements. For example: $(1, 2, 3)$ is a 3-tuple, and $(1, 2, 3) \neq (3, 2, 1)$ even though $\{1, 2, 3\} = \{3, 2, 1\}$ ⁷. 2-tuples are sometimes called a *pair*. If it is clear from context, it is a common convention to drop the number in front of the word tuple, i.e., if the writer consistently uses 2-tuples they might choose to just call them tuples.

Elements

As mentioned earlier, If something is inside a set, we call that an *element* or *object* of the set. We usually denote elements with lower-case letters (ex. a, b)⁸. To write “ a in A ” in mathematical notation, we use the symbol $\in$. For example:

$$a \in A$$

is read as “ a in A ” (or “ a is in A ” to sound more grammatically correct). If we want to say that “ a not in A ”, we add a slash across our symbol ($\notin$):

$$a \notin A$$

⁷In set theory, every mathematical object is represented as a set. This is no exception in n -tuples. A 2-tuple (a, b) would be $\{a, \{a, b\}\}$. A similar pattern follows for n -tuples

⁸Here’s another “beyond the scope of naive set theory” question: What are the possible elements that may exist? Question of this varieties lead to logicians like Butrard Russel to come up with *Russel’s Paradox*. This paradox will actually be covered in the book near the end (see ref:HERE). It’s written in such a way that you can skip to take a look at it if you’re curious

For example, $1 \in \{1, 2, 3\}$, but $4 \notin \{1, 2, 3\}$. If we have the empty set $\emptyset$ (i.e. $\{\}$), then it is always the case that nothing is inside it, i.e., $a \notin \emptyset$ for any a . Sets can also contain other sets, so $\{1\} \in \{\{1\}, 2, \{\text{cheesecake}, 6\}\}$. Note that $1 \notin \{\{1\}, 2, \{\text{cheesecake}, 6\}\}$ since the set $\{\{1\}, 2, \{\text{cheesecake}, 6\}\}$ doesn't contain "the element 1" but the "the set that contains the element 1".

0.1.2 Concept Check Is

$$\{1, 2, 3\} \in \{0, \{\{1, 2, 3\}\}, 4\}$$

Equality

The concept of equality was already used earlier based on our intuition. We will expand a bit further on how to think of equality in set theory, and we will distinguish two ways of looking at equality:

$$A = \{1, 2, 3\}$$

$$A := \{1, 2, 3\}$$

Starting with the first line, the symbol $=$ means *logically equal*. Being "logically equal" (which we will always abbreviate to saying equal⁹) means the two symbols represent the same thing. For example, the number one-half and two-quarters are equal:

$$\frac{1}{2} = \frac{2}{4}$$

even though the symbol $\frac{1}{2}$ and $\frac{2}{4}$ are different symbols. We would read $\frac{1}{2} = \frac{2}{4}$ as "one-half is equal to two-quarters". If we want to say that the two are not equal (i.e. the symbols represent different things¹⁰), we do the same convention as with $\in$ and put a slash across the symbol: $\frac{1}{2} \neq \frac{1}{3}$.

Example 0.1.3: Examples Of Equality

1. if we want to say two sets have the same elements, that they are each representing the same collection of objects, we write $A = B$
2. $\{1, 2, 3\} \neq \{2, 3, 4\}$. Even though both sets have the elements 2 and 3, they each contain a number the other doesn't have, and so they are not equal
3. Similarly, $\{1, 2\} \neq \{1, 2, 3\}$ since $3 \notin \{1, 2\}$
4. As seen in a elementary class in mathematics, $\frac{1}{2} = \frac{2}{4}$, but $\frac{1}{2} \neq \frac{1}{4}$ ^a

^aNotice that in all the previous examples, we were comparing sets, while in this example we're comparing numbers. In proper set theory, every object is actually a set. Thus, even comparing $\frac{1}{2}$ and $\frac{2}{4}$ will be a comparison with sets. We will actually construct $\mathbb{Q}$ and show how it's elements are sets in section ??

⁹This is another big set-theoretic question: what does it really mean for something to be equal? Bertrand Russel and Alfred Whitehead in their book *Principia Mathematica* famously took 379 pages before they defined equality. In modern set theory, two sets are equal if they contain the same elements

¹⁰This is another question beyond naïve set theory: Since all we're using are symbols that represent objects, where do the objects come from? One answer to this is some universal set with every possible thing as an element of it, but thinking about what exactly does it mean for a set to contain everything leads to interesting nuances (one of the problems of a set that contains everything is once again Russel's Paradox)

The symbol $:=$ means the same thing as $=$, that is, $:=$ is also logically equal. The difference between $:=$ and $=$ is that $:=$ is used to emphasize that the symbol on the left of $:=$ will now be used to represent the thing on the right of $:=$. This is most often used in definitions to make sure the reader understands that symbol will consistently be used to represent the same thing many times. Because of this, the symbol $:=$ is read as “define” – the statement $A := \{1, 2, 3\}$ can be read that “Define A to be the set $\{1, 2, 3\}$ ”.

Set-Builder Notation

Many times, we want to have a set of elements that satisfy only certain properties. For example, from the set of all deserts, I only want the ones that contain ice-cream, or from the set of all numbers I want only the prime numbers, or from the set of all months, I want the months that end in “ber”. The way we’ll capture this idea with sets is by using *set builder-notation* to build a new set given an old one. If S is a set and P is some sentence describing a property (ex. being an even number, or being a prime number, or representing “this desert has ice-cream”), then we’d write

$$A = \{s \in S \mid P\}$$

or

$$A = \{s \in S : P\}$$

and is read “s in S such that [s satisfies the property P]”. In the next section, we shall be more precise about what type of sentence can be used:

Example 0.1.4: Set Builder Notation

1. If

$$M = \{\text{January, February, March, April, May, June, July, August, September, October, November, December}\}$$

and $D = \{m \in M \mid m \text{ ends in “ber”}\}$, then

$$D = \{\text{September, October, November, December}\}$$

2. If $W = \{\text{hello, (words in other languages)}\}$, and $V = \{w \in W \mid w \text{ is in Vietnamese}\}$, then

$$V = \{\text{here}\}$$

3. if $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ and $P = \{n \in \mathbb{N} \mid n \text{ is a prime number}\}$ then

$$P = \{2, 3, 5, 7, 11, \dots\}$$

Note that $P' = \{n \in \mathbb{N} \mid \text{the only numbers that divide } n \text{ are } 1 \text{ and } n\}$ is the exact same set (i.e. $P = P'$). Different properties can produce the same sets.

4. if $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ represents the set of all non-negative numbers with no decimal places, the set

$$A = \{x \in \mathbb{N} \mid x = 2y, y \in \mathbb{N}\}$$

would represent all even numbers ^a, and can be read “ x in the natural numbers such that x can be twice any natural number”

^aThis notation has it’s limitations if you dig deeper in it. You can create some very interesting paradoxes (Russel’s paradox being one of them). We will not construct such contradictory cases in the book

0.1.1 Propositions and Predicates

We will now explore how to formulate questions and mathematical statements using mathematical language.

Intuitively, there is some understanding of the meaning of a sentence¹¹. If I say “I think blue is pretty” then in your mind some information got passed on. If we make a sentence that has a truth value associated to it (the sentence can either be true or false)¹², like “The number 2 is prime”, then we call such a sentence a *proposition*. If we have a sentence that takes in a variable that will make it depend on its truth value, for example “The number x is a prime number”, then this will be called a *predicate*. Propositions and predicates are at the heart of mathematical communication. We have already encountered predicates before when we used set-builder notations: the restriction *has to be a predicate*. For convenience, we assign a symbol to a proposition. For example, we can have P = “2 is a prime number”. Then P would be true. However, Q = “2 is not a prime number” is a false. For predicates, we shall usually assign a symbol and a variables, for example:

$$R(x) = “x \text{ is a prime number}”$$

where x may be any natural number, then this predicate truth-value (that is, whether it is true or false) depends on the value the variables x will take.

Example 0.1.5: Predicates

1. let $P(x) = “x \text{ is a month of the year}”$. Then

$$P(\text{September}) = \text{true} \quad P(\text{Wednesday}) = \text{false} \quad P(\text{Laughing}) = \text{false}$$

2. The variable x can also be other propositions. For example we can have $P(x) = “x \text{ is a dog}”$ and $Q(x) = “x \text{ is an animal}”$.
3. Propositions can be composed, for example consider $R(x) = “P(x), \text{ then } Q(x)”$ where $P(x), Q(x)$ are the propositions from the above. When is this proposition true?
4. Proposition can also take in multiple variables, for example

$$P(x, y) = (x < y)$$

Predicates of the form

$$\text{if } P(x) \text{ then } Q(x)$$

¹¹The nature of how we derive meaning from sentences is an interesting logical and philosophical debate that goes beyond the scope of this book. See Russel, Wittgenstein, -----, for more details

¹²Development in a field called *topoi theory* are trying to generalize our notion of proposition beyond true and false, an interesting developing field to look into

are some of the most common, allowing for relations being made between different concepts. The symbol used to symbolize an “if, then” relation is $\Rightarrow$, so

$$\text{If } P(x) \text{ then } (Q) \quad \text{becomes} \quad P(x) \Rightarrow Q(x)$$

Which is usually read “ $P(x)$ implies $Q(x)$ ”. A common mistake is to assume that if $P(x) \Rightarrow Q(x)$, then $Q(x) \Rightarrow P(x)$. This need not be the case. For example, define the following two predicates

$$A(x) = (x|2) \Rightarrow (x|4) \quad A'(x) = (x|4) \Rightarrow (x|2)$$

When is predicate $A(x)$ true? When is $A'(x)$ true? If it is true that $P(x) \Rightarrow Q(x)$ always implies $Q(x) \Rightarrow P(x)$ we use the symbol $P(x) \Leftrightarrow Q(x)$ and read $\Leftrightarrow$ as “if and only if” (sometimes abbreviated to iff).

Vacuous Truth

Let’s say we had two proposition $P(x, y) = x \Rightarrow y$, and let’s say that x was false, while y is true. What is the truth value of $P(x, y)$? For example, let’s say that

$$\begin{aligned} x &= \text{I’m walking in the rain} \\ y &= \text{I am wet} \end{aligned}$$

Then if you are *not* walking in the rain, but *are* wet, then what should be the value of $P(x, y)$? Perhaps more interesting is the following example:

1. Let’s say there are no cell-phones in a room. Then is the statement “all phone are turned off” true?
2. A famous such example is the following, is the statement “The king of France is not wearing pants” true or false if France does not have a king?

There are many solutions to this problem. The mathematically accepted solution to this problem is something called *vacuous truth*, which gives the following resolution:

$$(\text{False} \Rightarrow \text{True}) \text{ is True, and } (\text{False} \Rightarrow \text{False}) \text{ is also True}$$

The idea being that any “nonsense” statement is always “true” in the most trivial way. This interpretation of truth is not a unique one, but it is one that is very convenient as shall be seen by the reader.

Logical Quantifiers

We will now explore how to create more complicated properties by adding to our tool belt 6 “words” from which we can construct mathematical sentences:

Definition 0.1.6: Quantifiers And Connectives

We define the following 3 quantifiers to be used in propositions:

1. $\forall$: the for-all quantifier.
2. $\exists$: the *existence* quantifier
3. $\exists!$ the *unique existence* quantifier

and the following 4 connectives to be used for propositions

1. $\vee$: disjunction (or) connective. If P, Q are propositions, then $P \vee Q$ mean that if one of or both of P or Q is true, then the new proposition $P \vee Q$ is true.
2. $\wedge$: conjunction (and) connective. If P, Q are propositions, then $P \wedge Q$ mean that if both P and Q are true, then the new proposition $P \wedge Q$ is true. If one of them are false, then $P \wedge Q$ is false.
3. $\Rightarrow$: implication connective: $P \Rightarrow Q$ is false if P is true and Q is false. In other words, if P is true, then P implies Q means Q is true.
4. $\neg$ negation (not) connective. If P is a proposition, then if P is true then $\neg P$ is false, and similarly if P is false then $\neg P$ is true.

The keen reader may ask if there are any other quantifiers or connectives, and indeed there can be many more. However, it is in fact sufficient to define these connectives/quantifiers to completely describe any [first-order] mathematical sentence¹³. Using these symbols, can the reader translate some of the propositions earlier in the book using these symbols?

Recall that we have defined the symbol $\in$ earlier to specify an element belongs to a set. This is a special symbol which is neither a quantifier nor a connective. It is considered a special symbol that is part of the language of set theory.

The order of quantifiers is important. For example:

$$\forall x, \exists y, x < y$$

means that for every x , there exists a y that is greater than it, which is certainly true when working with numbers. However:

$$\exists x, \forall y, x < y$$

means that there exists an x that is smaller than all y , which is certainly not possible since we can always find a smaller number.

With these, we can define certain operations we can perform on sets (don't forget to add visuals):

¹³This would be explored in a course on logic

Definition 0.1.7: Set Operations

1. Define a subset to be:

$$A \subseteq B \Leftrightarrow (x \in A \Rightarrow x \in B)$$

Sometimes the symbol $A \subset B$ is also used. Since it is possible that $A \subset B$ $A = B$ (you can check this by the definition), the symbol $\subseteq$ will predominantly be used to emphasize this relation. If we want to make it clear that $A \neq B$, we write $A \subsetneq B$, we will write $A \subsetneq B$.

For any set S , we always have $\emptyset \subseteq S$ and $S \subseteq S$. With this notation, we also have that $A \subseteq B$ and $B \subseteq A \Rightarrow A = B$

2. Define the intersection of two sets to be:

$$A \cap B := \{x \mid x \in A \text{ and } x \in B\}$$

3. Define the union of two sets to be:

$$A \cup B := \{x \mid x \in A \text{ or } x \in B\}$$

if $A \cap B = \emptyset$, we say that A and B are *disjoint*

4. Define difference between sets to be:

$$A \setminus B := \{x \mid x \in A \text{ and } x \notin B\}$$

sometimes also written as $A - B$. If $T \subseteq S$, then we say that $S \setminus T$ is the *compliment* of T , usually written T^c .

Another very important definition

Definition 0.1.8: Product Of Sets

Let A and B be sets. Then

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

is the set of all 2-tuples (called just tuples for short). This construction can be extended indefinitely:

$$X_1 \times X_2 \times \cdots = \{(x_1, x_2, \dots) \mid x_1 \in X_1, x_2 \in X_2, \dots\}$$

This notation is also abbreviated as:

$$\prod_{i=1}^n X_i$$

For example:

Example 0.1.9: Product Of Sets

$$\{1, 2\} \times \{a, b\} = \{(1, a), (2, a), (1, b), (2, b)\}$$

and

$$\{a, b\} \times \{1, 2\} = \{(a, 1), (b, 1), (a, 2), (b, 2)\}$$

(A word on arbitrary products? i.e. $\prod_{i \in I} X_i$)**Exercise 0.1.1**

1. Let $\{E_n\}_{n=1}^{\infty}$ be a collection of sets indexed by the natural numbers. Define

$$\limsup\{E_n\}_{n=1}^{\infty} = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} E_n \quad \liminf\{E_n\}_{n=1}^{\infty} = \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} E_n$$

show that

$$\begin{aligned} \limsup\{E_n\}_{n=1}^{\infty} &= \{x \mid x \in E_n \text{ for infinitely many } n\} \\ \liminf\{E_n\}_{n=1}^{\infty} &= \{x \mid x \in E_n \text{ for all but finitely many } n\} \end{aligned}$$

2. (Demorgans La)

Naming the Numbers and Common Sets

With the basics of set down, we can introduce some of the most common sets and their notation

Definition 0.1.10: Common Sets

1. As we've already seen, if a set has no elements, the symbol $\emptyset$ represents the empty-set
2. If a set has a single element, we call that set a *singleton*. For example, $\{1\}$ and $\{\pi\}$ are singletons.
3. If X is a set, $P(X)$ is the set of all subsets of X ^a. In set builder notation, it can be denoted $P(X) = \{S \mid S \subseteq X\}$. For example, if $X = \{a, b, c\}$, then

$$P(X) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$

^aThe existence of this set is an axiom in ZFC**Definition 0.1.11: Sets Representing Numbers**

1. $\mathbb{N} := \{0, 1, 2, 3, \dots\}$ is the set of all *natural numbers* (some call it the *counting numbers* or *whole numbers*). Natural numbers are all positive numbers which do not have any decimal places. Sometimes, people define $\mathbb{N}$ to not have 0. We will always have $\mathbb{N}$

contain the number 0 (for reasons which will be clear much later in the book ^a). If we want only the [strictly] positive number ^b, we will write $\mathbb{N}_{>0}$.

2. $\mathbb{Z} := \{\dots - 3, -2, -1, 0, 1, 2, 3, \dots\}$ is the set of all *integers*. Integers are the natural numbers along with their negative counter-part. Sometimes, the natural numbers are called the non-negative integers.
3. $\mathbb{Q} := \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}$ is the set of all *rational numbers*. For example, $\frac{1}{2} \in \mathbb{Q}$. The symbol $\mathbb{Q}$ is used since numbers of the form $\frac{p}{q}$ are called *quotients*. Notice also that this is one of the instances where we use the convention that if there is a repeated element, we “squish” to a single element. If we did not take this convention, this set wouldn’t be defined, since there would be many repeated numbers (for example $\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$ and so forth)
4. $\mathbb{R}$ represents any number with any decimal expansion (ex. $\pi, e, 1, \frac{1}{3}$), i.e., all the numbers on the number line. If you’ve learned about Complex numbers or Quaternions in high school, these numbers are not real numbers, and will be defined and studied later in this book. The set $\mathbb{R}$ is called the *real numbers*. We can abuse notation (meaning use the notation in a way bends the rules a little) and write $\mathbb{R}$ as

$$\mathbb{R} := \{\dots D_3 D_2 D_1 . d_1 d_2 d_3 \dots \mid D_i, d_i \in \mathbb{Z}\}$$

where $\dots D_3 D_2 D_1 . d_1 d_2 d_3 \dots$ represents some arbitrary string of numbers put together where there can only be finitely many D_i ’s, but there can be infinitely many d_i ’s. In other word, the number

$$0.123123123123\dots$$

where 123 continues on forever on the right is in the set $\mathbb{R}$, but

$$\dots 123123123$$

where 123 continues on forever on the left is not in the set $\mathbb{R}$.

^aSee section ??

^bThere is no complete agreement on whether 0 is positive or negative: 0 is either both positive and negative or neither positive nor negative. It usually comes down to the purposes of the context. If we allow 0 to be a positive number, we sometimes say that all numbers larger than 0 are *strictly positive*

Some find it intuitive to think about $\mathbb{R}$ has having “no gaps”, which means we cannot perfectly split $\mathbb{R}$ into two using the less than and greater than operator ($<$ and $>$) without missing a number. For an example of a split, we can split $\mathbb{N}$ into $\mathbb{N}_{\frac{5}{2}+} = \{x \in \mathbb{N} \mid x > \frac{5}{2}\}$ and $\mathbb{N}_{\frac{5}{2}-} = \{x \in \mathbb{N} \mid x < \frac{5}{2}\}$, and we have successfully split $\mathbb{N}$ into two without missing a number (i.e. every natural number is either in $\mathbb{N}_{\frac{5}{2}+}$ or $\mathbb{N}_{\frac{5}{2}-}$). If we chose to split around 5, then neither the set $\mathbb{N}_{5+}$ nor the set $\mathbb{N}_{5-}$ would contain 5, meaning it does not split $\mathbb{N}$. With the Real numbers, there does not exist a perfect split. You might ask if $\mathbb{Q}$ has this same property, but we will prove that we can in fact always split it: We will show $\mathbb{Q}_{\sqrt{2}+} = \{x \in \mathbb{Q} \mid x > \sqrt{2}\}$ and $\mathbb{Q}_{\sqrt{2}-} = \{x \in \mathbb{Q} \mid x < \sqrt{2}\}$ splits $\mathbb{Q}$ (see ref:HERE).

The fact that $\mathbb{R}$ cannot be split into two using $>$ and $<$ is actually at the heart of many formal definitions of $\mathbb{R}$ which is covered in an Analysis I class. In this book, we will not need to think of $\mathbb{R}$

in this fashion, and so we'll stick with just using intuition for now. ¹⁴

<https://math.stackexchange.com/questions/2834876/convergence-of-geometric-series-with-r1>

0.1.2 Induction

Here. The whole $(p(x) \Rightarrow p(x+1)) \Rightarrow$ here this as the formal definition? + some intuition?

0.1.3 *Pigeon-hole Principle

(Comes up in algebraic number theory, so should add it here)

0.1.4 Relations and Equivalence Relations

What does it mean two things to be equal? There are actually many ways, depending on what you want to be “equal”. For example,

1. we may say that two people have the same birthday, even if they are not born in the same year.
2. [more examples her](#)

In this section, we shall explore how to capture this more general notion of equality.

Relations

We start of with the more general notion of *relation*. Many times when working with sets, some subsets can be thought of as having certain properties for example, we can split the set $\mathbb{Z}$ into even and odd numbers, or we have some numbers being bigger than others, or that some numbers are multiples of another. The following is this idea is formalized:

Definition 0.1.12: [Binary] Relation

Let X be a set. Then any collection of pairs (x, y) , $x, y \in X$ is called a [binary] *relation*. If $(a, b) \in R$, we will say that x is *related to* y . If $(a, b) \in R$, we will write aRb .

More generally, we can increase the number of times we take the product (i.e. $n > 2$) (ex. $X \times X \times \cdots X$). This would be called an *n-relation*.

We shall almost always think of relations

¹⁴Google *Dedekind Cut construction* or *Completion of Rational Numbers* if you're curious to see a formal definition of $\mathbb{R}$

Example 0.1.13: Relations

1. Take $\mathbb{Z}$ and define the set $<$ to be

$$(x, y) \in < \text{ if and only if } x \text{ is strictly less than } y$$

Then $(3, 4) \in <$, $(3, 3) \notin <$, and $(4, 3) \notin <$. If $(a, b) \in <$, we would usually write:

$$a < b$$

and if $(a, b) \notin <$, we would write

$$a \not< b$$

A similar relation is the $\leq$ relation, where we also allow for two numbers to be equal

2. Take $\mathbb{Z}$ and define the set R_{div} to be:

$$(x, y) \in R_{\text{div}} \text{ if and only if } x \text{ divides } y$$

Then $(2, 4) \in R_{\text{div}}$. What other elements are in this set? This set is usually denoted $|$ so that if a divides b then:

$$a|b$$

3. for a non-mathematical example, if we let X be the set of all humans, then we may define a relation “is the ancestor of”. Another similar but different relation is “is the parent of” where two humans are related is one is the parent of another.

A common relation to put on objects give a sense ordering:

Definition 0.1.14: Order

Let R be a relation on X . Then R is a *partial order* if it is:

1. **reflective:** For all $x \in X$, xRx (x always relates to itself)
2. **antisymmetric:** if xRy and yRx , then $x = y$ ^a
3. **Transitive:** if xRy and yRz , then xRz .

Furthermore, a partial order is a *total order* (or simply an *order*) if it is furthermore:

4. **Connected:** For each $x, y \in X$, either xRy or yRx

^aa relation is symmetric if xRy then yRx . Hence, antisymmetry means no elements can be symmetric with another elements besides itself

The canonical total order is $\leq$ on numbers, and the canonical partial order is $\subseteq$ on sets.

Example 0.1.15: Orders

1. Let $\mathbb{N}^2 = \mathbb{N} \times \mathbb{N}$. Give some examples on $\mathbb{N}^2$ of different ordering (Ex. dictionary, and others)

2. Take any dictionary. Then they are lexicographically ordered (more words here)
3. All numbers we introduced so far come have an order defined: $(\mathbb{N}, \leq)$, $(\mathbb{Z}, \leq)$, $(\mathbb{Q}, \leq)$, $(\mathbb{R}, \leq)$. This is the order define in high school (so $3 \leq 4$ or $3 \leq \pi$)
4. other cool example

Equivalence Relations

The most common type of relation that we shall see is the equivalence relation which generalizes the notion of equality:

Definition 0.1.16: Equivalence Relation

Let S be a set and $x, y, z \in S$ be elements of S . An *equivalence relation* on a set S is relation $\sim$ satisfying

1. **(reflective)** $x \sim x$ (i.e. every element must be related to itself)
2. **(symmetric)** if $x \sim y$ then $y \sim x$
3. **(Transitive)** if $x \sim y$ and $y \sim z$ then $x \sim z$

Example 0.1.17: Equivalence Relations

1. Equality ($=$) is the prototypical example of an equivalence relation. For example, If we have $a, b, c \in S$ with $aRb \Leftrightarrow a = b$, then
 - (a) $a = a$
 - (b) $a = b \Rightarrow b = a$
 - (c) If $a = b$ and $b = c$, then $a = c$.

In a ways, an equivalence relation is a generalization of equality.

2. Let X be any non-empty set. Then we may make all elements equivalent to each other. Similarly, we may make all elements only equivalent to themselves.
3. Let P be the set of all people. Check that “has the same birthday” is an equivalence relation
4. Two equivalence relations from high school that the reader may have encountered is with triangles. Check that being *similar* to and being *congruent* to are equivalence relations.
5. Define $\sim$ on $\mathbb{Z}$ to be $a \sim b$ if and only if $|a| = |b|$. Then $\sim$ is an equivalence relation.
6. Is $\leq$ an equivalence relation?
7. Let R be the empty relationship on any non-empty set. Is R an equivalence relation?

Given an equivalence relation $\sim$ on S , we may split S into subsets defined as:

$$(S/\sim) = \{T \subseteq S \mid x, y \in T \iff x \sim y\}$$

For example, if $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, and:

$$1 \sim 2 \quad 3 \sim 4 \sim 5 \sim 6 \quad 7 \sim 8 \sim 9$$

Then:

$$S / \sim = \{\{1, 2\}, \{3, 4, 5, 6\}, \{7, 8, 9\}\}$$

In fact, we may define an equivalence relation on a set S by breaking it up into disjoint sets whose union of elements would be S . This is called a partition:

Definition 0.1.18: Partition

Let S be a set. Then a *partition* of S is a collection of subsets of S such that

1. for each subset S_i, S_j , $S_i \cap S_j = \emptyset$
2. $\cup_i S_i = S$

each partition is called an *equivalence class*.

0.1.5 Functions

In mathematics, it is very desirable to relate one set to another set, or said differently how two sets may “interact”. For example, it may be desirable that to each positive natural number $\mathbb{N}_{>0}$, we associate to it the number of prime factors it has, for example:

$$\begin{aligned} 1 &\mapsto 1 \\ 2 &\mapsto 1 \\ 3 &\mapsto 1 \\ 4 &\mapsto 2 \\ 5 &\mapsto 1 \\ 6 &\mapsto 2 \\ &\vdots \\ 16 &\mapsto 4 \\ &\vdots \end{aligned}$$

Another more abstract relation we may ask for is given a set of names, say

$$\{\text{Alfred, Socrates, Buddha, Oscar Peterson}\}$$

the function returns a set containing some information about that person:

$$\begin{aligned} \text{Alfred} &\mapsto \{\text{Batman's Butler, Friend of Bruce}\} \\ \text{Socrates} &\mapsto \{\text{Philosopher, lived in Athens, Never wrote his work down}\} \\ \text{Buddha} &\mapsto \{\text{Real name Siddhartha Gautama, Starter of a Religion, Born in Lumbini}\} \\ \text{Oscar Peterson} &\mapsto \{\text{Greatest Jazz Pianist, Canadian, "Maharaja of the keyboard", Lived to 82}\} \end{aligned}$$

As a final example, we may imagine a every location of a map of the world being represented as an element in a set. Then we may relate a point on the map to the temperature at that location. Generalizing this pattern, given some set X , we may want to assign it an element from another set Y .

Definition 0.1.19: [set] Functions

Let X and Y be sets. Then a *function* is a subset $\Gamma(f) \subseteq \{(x, y) \mid x \in X, y \in Y\}$ such that

1. **Total** or **Serial** (left-total): For every $x \in X$, there exists an element $(x, y) \in \Gamma(f)$
2. **Functional** (right-unique): Every $x \in X$ is associated with one and only one $y \in Y$, that is, it cannot be that $(x, y), (x, y') \in \Gamma$, for $y \neq y'$ where $y, y' \in Y$

This is usually written as:

$$f : X \rightarrow Y \quad f(x) = y \quad \text{or} \quad X \xrightarrow{f} Y$$

where f is called the *function* and $\Gamma(f)$ is called the graph of the function and is defined to be:

$$\Gamma(f) = \{(x, y) \mid x \in X, y \in Y, f(x) = y\}$$

If $f : X \rightarrow Y$, X is called the *domain* and Y is called the *codomain*. The collection of all functions from two sets X, Y is denoted:

$$\text{Hom}_{\text{Set}}(X, Y) \quad \text{or} \quad Y^X \quad (a)$$

^aNotice this is a set operation. This particular notation shall be elaborated on with more category theory in section ref:HERE

Function are also sometime called maps or mappings. Sometimes, we want to define a function, but don't need to reference to it again (in computer science, this is called an *anonymous function*). In this case, we may use the following notation:

$$x \mapsto y$$

instead of $f(x) = y$. If $X = Y$ so that $f : X \rightarrow X$, then we may think of f as a relation which has the property of being functional and total. If more than one function with the same domain and codomain is defined (say f, g with domain X and codomain Y), then the declaration is often shorthand to $f, g : X \rightarrow Y$.

Example 0.1.20: Functions

1. The first three examples given are examples of functions, since every element in the domain get's mapped to a unique element in the codomain.
2. Some other perhaps more familiar functions to the reader are
 - (a) $x \mapsto e^x$, i.e. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^x$
 - (b) $\log : \mathbb{R}_{>0} \rightarrow \mathbb{R}, x \mapsto \log(x)$
 - (c) $p(x) : \mathbb{R} \rightarrow \mathbb{R}, p(x) = ax^2 + bx + c$, for $a, b, c \in \mathbb{R}$

3. Let X be an arbitrary non-empty set. Then there always exists at least one function from X to X , namely:

$$\text{id}_X : X \rightarrow X \quad \text{id}_X(x) = x$$

that is, id_X maps every element to itself.

Definition 0.1.21: Image and Restriction of Function

1. Let $S \subseteq X$ and $f : X \rightarrow Y$. Then:

$$f(S) := \{f(x) \mid x \in S\}$$

this is called the *image* of S . If $S = X$, then $f(X)$ is called the *image* or *range* of the function f , and is commonly denoted $\text{im}(f)$.

2. Let $S \subseteq X$. Then $f|_S$ is a function $f|_S : S \rightarrow Y$ such that

$$f|_S(x) = f(x)$$

that is, $f|_S$ is the *restriction* of f by S .

As a function is just some “syntactic sugar” to more easily represent the set Γ_f , two functions f, g are equal if $\Gamma_f = \Gamma_g$. In terms of function notation this means that f, g are equal if for all elements x in the domain:

$$f(x) = g(x)$$

An advantage of the left-total and right-unique property is that it is easy to compose functions together, that is it is easy to chain together the output of one function to be the input of the next:

Proposition 0.1.22: Composition of Functions

Let $f : X \rightarrow Y, g : Y \rightarrow Z, h : Z \rightarrow W$ be function. Define $(g \circ f)$ to be $(g \circ f)(x) = g(f(x))$. Then $(g \circ f)$ is a function and is called *g compose f*. The process is called *function composition*.

Proof :

Make sure that $g \circ f$ satisfies the properties of functions

Example 0.1.23: Function Composition

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$ and $g : \mathbb{R} \rightarrow \mathbb{R}^2$ be $g(x) = (x, 0)$. Then:

$$(g \circ f)(x) = (x^2, 0)$$

A function may also be composed with itself, given the domain and codomain match:

$$(f \circ f)(x) = x^4$$

2. Let $f : X \rightarrow Y$ be any function, id_X, id_Y the identity functions defined in example 0.1.20. Then:

$$f \circ \text{id}_X = f \quad \text{id}_Y \circ f = f$$

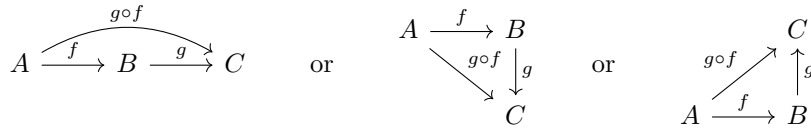
This chaining together is in a way “state-independent”: given three functions f, g, h , the function $(g \circ f)$ following by composing with h is equal to if we first just do f , then follow by taking the combined state $(h \circ g)$ ¹⁵:

Proposition 0.1.24: Associativity of Functions

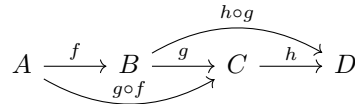
Let $f : X \rightarrow Y, g : Y \rightarrow Z, h : Z \rightarrow W$ be function. Then:

$$h \circ (g \circ f) = (h \circ g) \circ f$$

Pictorially, function composition may be represented by the following diagrams:



and associativity may be drawn as:



Proof :

This is a long and tedious proof that the reader should attempt at least once in their life.

These diagrams have the following special property: start on one of sets in the diagram (say A) and travel down the arrows to reach another set (say to D in the last diagram). Now start again at A and travel down to D via a different path in the diagram. Then notice that the compositions of functions agree with each other in the two paths. For example, take the following path:



Then the 1st red line is the composition $(h \circ g) \circ f$ and the 2nd red line is the composition $h \circ (g \circ f)$. Then by associativity, these two functions are equal

$$(h \circ g) \circ f = h \circ (g \circ f)$$

a diagram which satisfies this property for all paths is called a *commutative diagram*. These shall show up everywhere in this book, and mathematics more generally. They shall be further explored in section 0.4.

¹⁵can you think of an example where this is not the case in your everyday life? This would greatly help in understanding associativity

Injective, Surjective, and Bijective

There are two special types of functions that deserve to be pointed out: *injective* functions and *surjective* functions. These shall each represent some partial “reverse” of a function. Starting off with injectivity:

Definition 0.1.25: Injective

Let $f : X \rightarrow Y$ be a function. Then f is called *injective*, an *injection*, or *one-to-one* if the following condition holds: for all $a, b \in X$

$$f(a) = f(b) \Rightarrow a = b$$

To emphasize a function is injective, a hooked right arrow ($\hookrightarrow$) may be used like so:

$$f : X \hookrightarrow Y \quad X \xrightarrow{f} Y \quad X \hookrightarrow Y$$

Intuitively, being injective means we can embed the domain X directly into Y . As a consequence of this, we can always “recover” X in the following way

Proposition 0.1.26: Injective And Left Inverse

let f be an injective function. Then there exists a function $\ell f(X) \rightarrow X$ such that

$$\ell \circ f = \text{id}$$

that is $(\ell \circ f)(x) = \text{id}(x) = x$ is the identity function.

Proof :

Define $\ell : Y \rightarrow X$, $\ell(y) = \ell(f(x)) = x$. Since f is injective, this is a well-defined function (if it were not injective, then $\ell(f(x))$ would be ambiguous). Then by definition $(\ell \circ f)(x) = \ell(f(x)) = x$, as we sought to show.

Notice that the inverse function ℓ inverted the function *after* applying f . What if there is some function $r : Y \rightarrow X$ such that f reverses it? The following shows when this is the case:

Definition 0.1.27: Surjective

Let $f : X \rightarrow Y$ be a function. Then f is *surjective*, a *surjection*, or *onto* if for all $y \in Y$, there exists an x such that $f(x) = y$, i.e. $f(X) = Y$. To emphasize a function is surjective, a double-headed right arrow ($\twoheadrightarrow$) may be used like so:

$$f : X \twoheadrightarrow Y \quad X \xrightarrow{f} Y \quad X \twoheadrightarrow Y$$

Intuitively, being surjective means that Y is “represented” by the elements of X , i.e., every element y has one or many elements of x associated to it, which is why another word for a function to be surjective is that it is “onto”. As a consequence of this, we can pullback any element $y \in Y$ to some representative $x \in X$ such that $f(x) = y$. We can capture this intuition in the following way:

Proposition 0.1.28: Surjective And Right Inverse

Let $f : X \rightarrow Y$ be a surjective function. Then there exists a (and in fact more generally many) $r : Y \rightarrow X$ such that

$$r \circ f = \text{id}$$

that is $(f \circ r)(x) = \text{id}(x) = x$ is the identity function.

Proof :

Define $r : Y \rightarrow X$ by mapping each $y \in Y$ to some element $x \in X$ such that $f(x) = y$ (such an element exists since f is surjective). Then by definition for any $y \in Y$, $(f \circ r)(y) = f(r(y)) = f(x) = y$ as we sought to show.

Another perhaps more intuitive way of seeing how surjections make every y “represented” by X is by seeing that every surjection defines an equivalence class:

Proposition 0.1.29: Surjective And Equivalence Class

Let $f : X \rightarrow Y$ be a surjective function. Then we can define an equivalence class $\sim_f$ on X by defining:

$$a \sim_f b \quad \Leftrightarrow \quad f(a) = f(b)$$

If $[x] \in X / \sim_f$ is an equivalence class, such that $f(x) = y$, then $[x]$ is called the *fiber* over y .

Proof :

1. For each $a \in X$, $f(a) = f(a)$ by right-uniqueness, hence $a \sim_f a$
2. If $a \sim_f b$ for $a, b \in X$, then $f(a) = f(b)$. By symmetry of the equivalence relation $=$, $f(b) = f(a)$, and so $b \sim_f a$
3. Can the reader show that if $a \sim_f b$ and $b \sim_f c$, then $a \sim_f c$?

If f is not surjective, this equivalence relation may still be defined on the image of f . If f is injective, then each equivalence class has size 1. Many functions that are present in mathematics are both injective and surjective; to give a simple example:

$$f : \{1, 2, 3\} \rightarrow \{a, b, c\}, \quad f(n) = \begin{cases} a & n = 1 \\ b & n = 2 \\ c & n = 3 \end{cases} \quad (1)$$

Sch functions are given a name:

Definition 0.1.30: Bijective

Let $f : X \rightarrow Y$ be a function. Then f is said to be *bijective* or *isomorphic in **Set*** if f is injective and surjective. If X and Y are bijective, then we may write:

$$X \cong_{\mathbf{Set}} Y \quad X \cong Y \quad X \xrightarrow{\sim} Y$$

The intuition of bijective functions can combine the intuition behind injective and surjective functions, however they may equivalently be seen as formally defining a notion of “size”. As seen in equation 1, two sets that are bijective have the same size. The issue becomes more subtle when the domain/codomain are infinite:

Example 0.1.31: Size Of Infinite Sets

Define $2\mathbb{N} = \{2n \mid n \in \mathbb{N}\}$ and $f : \mathbb{N} \rightarrow 2\mathbb{N}$ by:

$$f(n) = 2n$$

Show that f is bijective.

This gives rise to the following definition:

Definition 0.1.32: Cardinality

Let X, Y be sets. Then X is said to have the same *cardinality* as Y if there exists a bijective function $f : X \rightarrow Y$. If X, Y have the same cardinality, they are said to have the same size.

Another important property of bijective functions is that they give rise to a *unique* inverse:

Proposition 0.1.33: Bijectivity And Unique Inverse

Let $f : X \rightarrow Y$ be a bijective function. Then there exists a unique function g such that

$$g \circ f = \text{id}_X \quad f \circ g = \text{id}_Y$$

where $\text{id}_X : X \rightarrow X$ and $\text{id}_Y : Y \rightarrow Y$ are the identity maps on the respective sets. Often, the subscript on the identity maps are dropped and the following abuse of notation is carried out:

$$g \circ f = f \circ g = \text{id}$$

The function g is often denoted f^{-1}

Proof :

Since f is surjective, there exists an inverse function $g : Y \rightarrow X$ such that $f \circ g = \text{id}_Y$. Since f is injective, there can only be one such function (there is only element in the fiber above each $y \in Y$). Furthermore, $\text{im}(f) = Y$, hence g satisfies the condition $g \circ f = \text{id}_X$, as we sought to show.

0.1.34 Remark: inverse notation abuse The symbol f^{-1} has one other of notations that the reader should be aware, in particular this symbol may represent one of two things depending on the context. Given $f : X \rightarrow Y$:

1. if f is bijective, f^{-1} represents the unique inverse function
2. $f^{-1} : P(Y) \rightarrow P(X)$ is the function

$$f^{-1}(T) = \{S \in P(X) \mid S = \{x \in X \mid f(x) = y\}\}$$

the images of f^{-1} are called the *pre-images* of f . This notation is particularly prevalent in topology where this the pre-image function takes center stage through continuous functions, however it will also have many applications in algebra.

Given any function $f : X \rightarrow Y$, there are certain natural functions that we may define that allows us to decompose f as a composition of injective and surjective functions with an intermediary bijective function

Proposition 0.1.35: Function Decomposition

Let $f : X \rightarrow Y$ be a function. Then there exists functions such that

$$X \xrightarrow{\quad} X / \sim_f \xrightarrow{\quad} \text{im}(f) \xrightarrow{\quad} Y$$

f

where the 1st function is surjective, the 2nd bijective, and the 3rd injective

Proof :

Define $\pi_f : X \rightarrow X / \sim_f$ by

$$\pi_f(x) = [x]_f$$

that is, π_f maps x to it's equivalence class. Next, define $F : X / \sim_f \rightarrow \text{im}(f)$ by

$$F([x]) = y$$

where $y \in Y$ is chosen so that $f(x) = y$. Finally, define $\iota : \text{im}(f) \rightarrow Y$ by

$$\iota(y) = y$$

that is, ι maps y to itself (since $\text{im}(f) \subseteq Y$). The reader should verify that these maps have the desired properties.

This proposition gives an intriguing relation between equivalence relations and functions, namely any function may be “represented” as an equivalence relation, and equivalently any equivalence relation may be “represented” by a function.

Using functions, we may define the notion of a multiset:

Definition 0.1.36: Multiset

Let X be a set. Then a *multiset* given by elements of X is a function:

$$m : X \rightarrow \mathbb{N}$$

which assigns each element $x \in X$ a multiplicity.

Example 0.1.37: Multiset

Let $X = \{a, b, c\}$ and:

$$m(a) = 3 \quad m(b) = 1 \quad m(c) = 0$$

Then m defines a multiset. Commonly, this set is written as:

$$\{a, a, a, b\}$$

As practice, define another multiset n and show how to define the notion of inclusion, union, intersection, and so forth.

As a closing remark, it is a common convention in the study of Analysis (one of the major studies of mathematics) that if we have a function $f : X \rightarrow Y$, we might abuse notation and use a subset of X as the actual domain. For example, $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{1}{x}$ is not a function as we've defined the concept, since $f(0) = \frac{1}{0}$ is not defined¹⁶. If we were to re-write this to be a function, we'd write $f : \mathbb{R} - \{0\} \rightarrow \mathbb{R}$. However, it is often implied in analysis that we will restrict the domain to the points for which f is defined.

Exercise 0.1.2

1. Let $f : X \rightarrow Y$ and take $A \subseteq X$, $B \subseteq Y$. Show that $f^{-1}(A) \subseteq B \Leftrightarrow A \subseteq f(B)$
2. Let $f : X \rightarrow Y$. Then is it true that $f(A \cup B) = f(A) \cup f(B)$? What about $f(A \cap B) = f(A) \cap f(B)$? What if there were more unions or intersections?
3. Let $f : X \rightarrow Y$, and define $f^{-1} : P(Y) \rightarrow P(X)$, $f^{-1}(Y) = \{x \in X \mid f(x) \in Y\}$. Is $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$? What about $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$? What if there were more unions or intersections?

(In a way, equivalence relations capture some idea of “projecting”. If you have some set S and you want to put together all equivalent objects, where equivalence here can be thought as “similar enough so that we can define the concept of equality”

4. (partitioning property. Like creating the “new objects” where we ignore the “details” that is “consumed” by the equivalence relation)
5. (natural projection?? Give $\pi : G \rightarrow \{0\}$ a natural projection that always exists?)

¹⁶This would be covered rigorously in a calculus or analysis class

0.1.6 Adding Structures to Sets

So far, we have been extensively working with sets and functions. Sets by themselves have no notion of size, order, arithmetic operations, geometry, among others possible properties. Many interesting subjects in mathematics deal with these additional concepts. For this reason, mathematicians often “augment”, so to speak, sets to have some extra structure:

Example 0.1.38: Extra Structure

1. (Order Structure). A set $(A, \leq)$ with an order $\leq$ is called an *ordered set*. Depending on the type of order given (ex. partial order, total order, well order etc.), the set would be called the appropriate order (ex. partially ordered set, totally ordered set, well ordered set, etc.)
2. (Algebraic structure) The set $\mathbb{Z}$ on its own is only numbers (and so is $\mathbb{N}, \mathbb{Q}, \mathbb{R}$). If we add a function $+$: $\mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$, where $+$ acts exactly like the familiar addition we have learnt earlier in life, then $(\mathbb{Z}, +)$ now also has addition! We can add more functions to sets, for example $(\mathbb{Z}, +, \cdot)$ where $\cdot$: $\mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ does the usual multiplication. If we augment a set with function(s), we call this an *algebraic structure*. This whole book will be dedicated to the study of algebraic structures.
3. (Topological Structures). Another common augmentation of a set is something called a topology. Given a set A , a set $\mathcal{T}_A$ is a set of subsets of A that follows three rules:
 - (a) $\emptyset \in \mathcal{T}_A$ and $A \in \mathcal{T}_A$
 - (b) if $\{U_i\}_{i \in I} \subseteq \mathcal{T}_A$ is some arbitrary subset of $\mathcal{T}_A$ indexed by I , then $\bigcup_{i \in I} U_i \in \mathcal{T}_A$ (i.e. an arbitrary union of sets in $\mathcal{T}_A$ is in $\mathcal{T}_A$)
 - (c) if $U_1, U_2, \dots, U_n \in \mathcal{T}_A$ is a finite subset of $\mathcal{T}_A$, then $\bigcap_i U_i \in \mathcal{T}_A$ (i.e. a finite intersections of sets in $\mathcal{T}_A$ is in $\mathcal{T}_A$)

Then $(A, \mathcal{T}_A)$ is called a *topology*. If a set is augmented with a sets of sets that follow these rules, it's called a *topological structure*. Topology is the natural space in which much of Real and Complex Analysis, as well as [non-classical] geometry takes place. It is the “coarsest” structure we add to sets to give sets some notion of locality.

4. (Measure Structures). Similar to a topological structure, a measure structure is one where
 - (a) $\emptyset \in \mathcal{T}_A$ and $A \in \mathcal{T}_A$
 - (b) if $U_1, U_2, \dots, U_n, \dots \in \mathcal{T}_A$, then $\bigcup_i U_i \in \mathcal{T}_A$ (i.e. a countable union of sets in $\mathcal{T}_A$ is in $\mathcal{T}_A$)
 - (c) if $U_1, U_2, \dots, U_n, \dots \in \mathcal{T}_A$, then $\bigcap_i U_i \in \mathcal{T}_A$ (i.e. a countable intersections of sets in $\mathcal{T}_A$ is in $\mathcal{T}_A$)

This structure is central to generalizing the notion of “size” for general sets and is the backbone defining the integral^a.

5. (Differential structure) If you have done calculus, then you are familiar with the notion of being differentiable. There is a field of study in which we can call “sets” differentiable. Given a set S , there is a collection of functions called an *atlas* (say, $\mathcal{A}$ represents this collection) which gives a set S a “differentiable structure”, and is usually denoted $(S, \mathcal{A})$. The details of what the atlas $\mathcal{A}$ contains is left for a course in differential geometry or diffeology.

6. *(Scheme structure) This will be a very important structure on a topological space that will dominate the readers who will study algebraic geometry. A topological space represents some idea of locality on a set. A *sheaf* represents having data on each (open) set in a topological space in such a way that we can use local data to glue together global data (think of solving a problem by breaking it down into smaller problems and knowing you can glue it back together).
7. *(Model) In section 0.1.1, we have shown how to make logical statements given sets. The full study how sets (usually augmented with other structures) may have statements which are true or provable is the domain of *model theory*. Usually, a set M along with some collection of functions and relations is augmented in such a way to be able to study the properties of logical statements. This would be covered in a class on first order logic or model theory more generally.

^aIn particular, the Lebesgue integral

This textbook is dedicated to the study of algebraic structures. It will sometimes be the case that we encounter ordered structures¹⁷, and only near the end of the book do we encounter topological structures (p. ??) where it is assumed the reader has taken a course covering point-set topology¹⁸. The reason the eventual mixing of structure is because different domains of mathematics start motivating each other in further exploration. For example, many geometers and algebraists found parallels between their fields, and hence algebraic geometry was born. Similarly, it was found that algebra can be used to solve topological problems, and hence algebraic topology was found. (A word on geometric group theory and something like LCH groups or Amenable groups?)

0.1.7 *Cardinality

(Here in there in the book, cardinality will come up. For ex, in proof that an injective map from $X \rightarrow X$ where X is finite is surjective, or when we figure out the algebraic numbers are countable, and other results later in the book)

(classical diagonalization proof?)

(as exercise, do $\{0, 1\}^{\mathbb{N}}$ is bijective to $[0, 1]$ by thinking of binary representation?)

¹⁷see section ??

¹⁸Differentiable structure's will be mentioned to motivate the concept of schemes which are based off of manifolds

0.2 Number Theory

Numbers are at the heart of mathematics and algebra. In this section, the basic properties of numbers will be explored. Some history of these concepts will be given as well to give a sense of why we call these special sets numbers.

Numbers started with our intuitive ability to understand discrete quantities (ex. 5 rocks, 2 people). In the past, it was contentious whether 0 was a number (the idea being that numbers are supposed to represent a real quantity), however it is now accepted that 0 represents the notion of “nothingness”, “emptiness”, or “inaction”. Ratios of numbers (ex. $3/4$) became important when it was important to be precise on what part of a whole object we want (ex. half the money). Negative numbers became important when considering other peoples debts (you owe 2 sheep, you “have” -2 sheep). In the times of at least the Greeks, it was discovered that there are certain numbers that are irrational (ex. $\sqrt{2}$ cannot be written as $\frac{a}{b}$, where a, b are natural numbers); the square root of any prime number shown to be irrational. The square-root of negative numbers remained undefined for awhile until NAME discovered that they are useful in finding roots of polynomials of degree 3, and it wasn’t till Euler that square-roots of negative numbers got the symbol $i := \sqrt{-1}$. It wasn’t till later that it was understood there were many more irrational numbers than rational numbers; the collection of all rational and irrational numbers form the real numbers. The collection of all numbers $a + bi$, $a, b \in \mathbb{R}$ is known as the *complex numbers*. For reasons that shall be explored later in the book¹⁹, any further addition to the number system beyond the complex numbers start to lose properties of numbers, and so we usually consider the following sets as “numbers” (ex. $ab \neq ba$ or $(ab)c \neq a(bc)$):

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

We shall heavily explore the properties of all of these number systems, with the notable exception of $\mathbb{R}$. This is because the real numbers are in many ways the beginning of the exploration of “taming infinities” and start moving beyond algebra. In fact, we shall even show in section ?? that it is impossible to construct $\mathbb{R}$ in an algebraic way.

0.2.1 GCD and Euclidean Algorithm

Definition 0.2.1: Divisibility

Let a, b be numbers. Then we say that a *divides* b , or b is *divisible by* a if there exists a number k such that

$$ak = b$$

Note that divisibility depends on the set of numbers you are working, In $\mathbb{N}$, 3 is not divisible by 5, however in $\mathbb{Q}$, 3 is divisible by 5, namely let $k = 3/5$.

Example 0.2.2: Divisibility Practice

1. Show that if $x|n_1$ and $x|n_2$, then $x|an_1 + bn_2$ for any $a, b \in \mathbb{Z}$.
2. Show that if $a|b$ and $b|c$ then $a|c$

¹⁹ref:HERE for details

From now on in this section, we shall limit ourselves to divisibility in $\mathbb{Z}$ ²⁰.

One special result about integers is that every number has a prime factorization:

Proposition 0.2.3: Prime Factorization Of Natural Numbers

Let $n \in \mathbb{N}$. Then there exists distinct prime numbers $p_1, p_2, \dots, p_k$ and natural numbers $n_1, n_2, \dots, n_k$ such that n can be uniquely broken down into:

$$n = p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$$

Proof :

exercise (hint: use induction)

This definition can be extended to $\mathbb{Z}$, except we lose uniqueness, the decomposition is unique up to a multiple of -1 . This isn't too much of a concern, since -1 has the special property of having a multiplicative inverse, that is we may multiply -1 by a number (in this case, itself) to get to 1:

$$-1 \cdot -1 = 1$$

no other integer (besides trivially 1) can be multiplied by another integer to get to 1. Since all numbers can be broken down into multiples of primes, we can define a way to compare how similar two numbers are:

Definition 0.2.4: Greatest Common Divisor

Let $a, b \in \mathbb{Z}$ be two non-zero integers. Then the *greatest common divisor* between these two numbers is the greatest number n such that n divides a and n divides b .

The common notation for this fact is that:

$$n|a \quad n|b$$

Similarly, we have the dual concept:

Definition 0.2.5: Least Common Multiple

Let $a, b \in \mathbb{Z}$ be non-zero integers. Then the *least common multiple* is the smallest integer m such that

$$a|m \quad b|m$$

Example 0.2.6: Greatest Common Divisor

1. $\gcd(10, 5) = 5$
2. $\gcd(100, 90) = 10$
3. $\gcd(45, 49) = 1$
4. If p, q are distinct prime numbers, then $\gcd(p, q) = 1$

²⁰In chapter ??, we shall re-define divisibility in a more general context

It is natural that two prime numbers have 1 as their greatest common divisor for they have no factor in common. However, it is not necessary for numbers to be prime for their gcd to be 1. We give a name to numbers with no common factors:

Definition 0.2.7: Coprime Numbers

Let $a, b \in \mathbb{N}_{>0}$. Then if $\gcd(a, b) = 1$, a and b are said to be *coprime*.

We may find the greatest common multiple by finding the prime factorization, however this is in general very inefficient on large numbers. These values can be found in an algorithmic manner relying on the fact that numbers have a notion of size²¹

Theorem 0.2.8: Euclidean Algorithm

Given two natural numbers a, b we can find their greatest common divisor by the following technic:

1. Let $a > b$. Put them into the equation $a = bk_1 + r_1$, where k is the amount of times b fits into a , and r_1 is the remainder.
2. Once you find these values, all but k shift over to the left, and you repeat the process to find r_2 :

$$\begin{aligned} b &= r_1k_2 + r_2 \\ r_1 &= r_2k_3 + r_3 \\ r_2 &= r_3k_4 + r_4 \end{aligned}$$

3. The algorithm continues until $r_n = 0$ (which it will reach since a, b where finite). The final non-zero remainder r_{n-1} is the greatest common divisor.

Proof :
exercise

As an example:

Example 0.2.9: Euclidean Algorithm

Let $a = 20$, $b = 13$. Then:

$$\begin{aligned} 20 &= 1 \cdot 13 + 7 \\ 13 &= 1 \cdot 7 + 6 \\ 7 &= 1 \cdot 6 + 1 \\ 6 &= 6 \cdot 6 + 0 \end{aligned}$$

²¹We shall see in chapter ?? that not all systems in which we can define a GCD will allow for a euclidean algorithm

so $\gcd(20, 13) = 1$, (the number above 0 in the last step). To find the $ax + by = 1$ formula, we reverse the process by converting the equation above in the following form:

$$7 - 6 = 1$$

so, put the remainder on the right hand side. Then, isolate the next equation in in the appropriate ways to substitute into this one: $7 = 13 - 6$, $6 = 13 - 7$. Since $6 = 13 - 7$, $7 = 13 - (13 - 7) = 7$, so

$$7 - (13 - 7) = 1$$

then, we repeat the same process with the step above: $7 = 20 - 13$, so:

$$20 - 13 - (13 - (20 - 13)) = 1$$

implying, we get

$$2 \cdot 20 - 3 \cdot 13 = 1 \quad (40 - 39 = 1)$$

giving us our desired equation.

Using the euclidean algorithms, we may always find the solution to the following interesting linear equation:

Theorem 0.2.10: Extended Euclidean Algorithm (Bézout's Identity)

Let $a, b \in \mathbb{Z}$. Then there exists integers $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a, b)$$

Proof :

exercise (reverse the process)

Example 0.2.11: Bézout's Identity

Recall that $\gcd(20, 13) = 1$. To find the $ax + by = 1$ formula, we reverse the process by converting the equation above in the following form:

$$7 - 6 = 1$$

so, put the remainder on the right hand side. Then, isolate the next equation in in the appropriate ways to substitute into this one: $7 = 13 - 6$, $6 = 13 - 7$. Since $6 = 13 - 7$, $7 = 13 - (13 - 7) = 7$, so

$$7 - (13 - 7) = 1$$

then, we repeat the same process with the step above: $7 = 20 - 13$, so:

$$20 - 13 - (13 - (20 - 13)) = 1$$

implying, we get

$$2 \cdot 20 - 3 \cdot 13 = 1 \quad (40 - 39 = 1)$$

giving us our desired equation.

Try doing the same with 220 and 38.

Bézout's Identity has the following neat interpretation: primes are fundamentally a multiplicative decomposition of numbers: any n is equal to $p_1^{n_1} \cdots p_k^{n_k}$ for appropriate primes p_i and powers n_i . However, they do have an additive relation between them, the Bézout identity. If we have two primes, we can (up to appropriate constants) add them to get to the multiplicative identity 1. In this way, we have a property of primes given via addition.

(should I add other number theory identities? Are there any other useful ones for this book? Perhaps just as exercises? They might be useful way later in algebraic number theory)

Exercise 0.2.1

1. Let $p, q, r \in \mathbb{N}$ be prime numbers greater than 3. Show that 3 divides $p^2 + q^2 + r^2$

0.2.2 Modular Arithmetic's

When counting the hours on a clock-face, we start with 12 o'clock, then add an hour to make it 1 o'clock, then another hour to make it 2 o'clock, until we circle back to 12 o'clock again, at which point we start counting again to 1 o'clock. For the rest of this section, we will say that 12 o'clock is 0 o'clock for reasons that would be explained soon. Notice how the arithmetic's of the clock different from the usual number system where for each $x \in \mathbb{N}$, there exists an n such that $n > x$. Hours on a clock could add to be zero, for example:

$$7 + 5 = 0$$

In this section, we rigorously define this notion

Definition 0.2.12: Modular Arithmetic's

Let $n \in \mathbb{N}_{>0}$. Define the equivalent relation on $\mathbb{Z}$:

$$a \sim_n b \iff a - b = kn \text{ for some } k \in \mathbb{Z}$$

The collection of these equivalence classes is denoted $\mathbb{Z}/n\mathbb{Z}^a$ and is called the *integers modulo n* . The number n is called the *modulus* and if $a \sim_n b$, we say that *a is congruent to b modulo n* . If $a \sim_n b$, then it is denoted

$$a \equiv b \pmod{n} \quad \text{or} \quad a \equiv_n b$$

The equivalence classes may be denoted as:

$$[a], [a]_n, \bar{a}, \bar{a}_n$$

where the n is dropped if the context permits it.

^aThis notation shall be elucidated on in chapter ??

Intuitively, this equivalence relation says that the difference of two numbers is a multiple of n . When $n = 12$, then this is the usual clock arithmetic's. When n is any other number, the reader may imagine a clock-face with n digits.

Naturally, it has to be proven that $\sim_n$ is indeed an equivalence relation:

Proposition 0.2.13: Modular Arithmetic's Well-defined

Let $\sim_n$ be defined as in definition 0.2.12. Then $\sim_n$ is indeed an equivalence relation

Proof :

Let $x, y, z \in \mathbb{Z}$ and let $n \in \mathbb{N}_{>0}$ be any positive natural number.

1. $x \equiv x \pmod n$ since $x - x = 0 = 0 \cdot n$
2. If $x \equiv y \pmod n$ then there exists a k st $x - y = kn$. Then $y - x = -kn = (-k)n$. Then by definition $y \equiv x \pmod n$
3. If $x \equiv y \pmod n$ and $y \equiv z \pmod n$, then $x - y = kn$ and $y - z = \ell n$ for some $k, \ell \in \mathbb{Z}$. Adding these two equations, we get:

$$x - z = x - y + y - z = (k + \ell)n$$

and so $x \equiv z \pmod n$

Since all the conditions of an equivalence relation is satisfied, $\sim_n$ is indeed an equivalence relation, as we sought to show.

Example 0.2.14: Modular Arithmetic's

Verify the following:

1. $35 \equiv 11 \pmod{12}$
2. $2 \equiv -3 \pmod{5}$
3. If $a \equiv b \pmod n$, then $a + t \equiv b + t \pmod n$ for any $t \in \mathbb{Z}$
4. If $a \equiv b \pmod n$, then $a^t \equiv b^t \pmod n$ for any $t \in \mathbb{N}$

Proposition 0.2.15: Properties Of Modular Arithmetic's

Let $n \in \mathbb{N}_{>0}$ and $a_1 \equiv a_2 \pmod n$, $b_1 \equiv b_2 \pmod n$. Then:

1. $a_1 + b_1 \equiv a_2 + b_2 \pmod n$
2. $a_1 - b_1 \equiv a_2 - b_2 \pmod n$
3. $a_1 b_1 \equiv a_2 b_2 \pmod n$

Furthermore:

1. $a_1 + b_1 \pmod n = a_1 \pmod n + b_1 \pmod n$ (in different notation, $\overline{a_1 + b_1} = \overline{a_1} + \overline{b_1}$)
2. $a_1 b_1 \pmod n = (a_1 \pmod n)(b_1 \pmod n)$ (in different notation, $\overline{a_1 b_1} = \overline{a_1} \overline{b_1}$)

Proof :

This is a good exercise

Proposition 0.2.16: Cancellation Properties Of Modular Arithmetic's

Let $n \in \mathbb{N}_{>0}$. Then:

1. if $a + k \equiv b + k \pmod{n}$ for any integer $k \in \mathbb{Z}$, then $a \equiv b \pmod{n}$
2. if $ka \equiv kb \pmod{n}$ for k coprime with n ($\gcd(k, n) = 1$) then $a \equiv b \pmod{n}$
3. if $ka \equiv kb \pmod{kn}$ for some $\mathbb{N}_{>0}$, then $a \equiv b \pmod{n}$

Proof :

good exercises

Important Results

The following are important properties of the integers modulo n :

Theorem 0.2.17: Fermat's Little Theorem

Let p be prime and $a \in \mathbb{N}_{>0}$ such that $p \nmid a$. Then

$$a^p \equiv a \pmod{p}$$

Proof :

First, we can assume that $0 \leq a \leq p - 1$, for we may always reduce a due to the properties of modular arithmetic's (hint: use the multiplicative properties of modular arithmetic's). Next, we may reduce this assertion to

$$a^{p-1} \equiv 1 \pmod{p}$$

For, if the above hold, then multiplying both sides by a gives the desired result.

Start by list out the first $p - 1$ positive multiples of a :

$$a, 2a, 3a, \dots, (p-1)a \tag{2}$$

I first claim that no two of these are equal mod p . For the sake of contradiction, let's say two were so that $ra \equiv sa \pmod{p}$ for some $1 \leq r, s \leq p - 1$ where $r \neq s$. Then since p is prime and $a > 0$, $\gcd(a, p) = 1$ and hence by proposition 0.2.15:

$$r \equiv s \pmod{p}$$

Since $0 \leq r, s \leq p - 1$, this implies $r = s$; contradiction our earlier claim. Hence, each term in list (2) is congruent (modulo p) to a unique number $1, \dots, p - 1$.

Now, multiply the terms in list (2). Then by proposition 0.2.15:

$$(1a)(2a) \cdots ((p-1)a) \equiv (1)(2) \cdots (p-1) \pmod{p}$$

which, when simplifying by combining terms, is:

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}$$

Since p is prime, each number $1 \leq k \leq p-1$ is coprime to p , and hence by repeated application of proposition 0.2.16:

$$a^{p-1} \equiv 1 \pmod{p}$$

but now, we may simply multiply both sides by a to get $a^p \equiv a \pmod{p}$, as we sought to show.

The reader may ask if this can be generalized in some way to modulo n . The key would be to generalize the step in which each number $1 \leq k \leq p-1$ is coprime with n , and indeed if $\varphi(n)$ represents the number of numbers that are coprime with n , then the identity can be generalized to:

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

This proof is best in the context of group theory, and so shall be left alone till then, however the persistent reader has all they need to attempt to directly prove this result.

(There are more cool proofs, but many rely on group theory or field theory. I'd like to include them in certain sections of the book: [here](#))

Exercise 0.2.2

1. Let $a \equiv b \pmod{n}$. If $k \mid a, b$, is it the case that $(a/k) \equiv (b/k) \pmod{n}$?

0.2.3 Counting Problems

(Also introduce the binomial theorem?) (somehow introduce the binomial theorem. Something else too?)

0.3 Graphs

mention how this is section is not necessary to read for groups, but it will come back up when working with particular perspective on groups. It can be revisited once you make it there (perhaps make the graph intuition of groups a series of exercises?)

(See this link: (commented for compiling purposes))

We first introduce a more concrete definition of a graph in order to be able to work with it unambiguously

Definition 0.3.1: Graph

Let $V = \{v_1, v_2, \dots, v_n, \dots\}$ be a set of elements representing the *vertices* (also called *nodes* or *points*). Let

$$E = \{\{v_i, v_j\} \mid v_i, v_j \in V\}$$

be the set representing the *edges* of the graph (where there need not be a set $\{v_i, v_j\}$ for every pair of vertices in V). Then $\mathfrak{G} = (V, E)$ is called a *graph*.

If the edges are *pairs* instead of sets, then we call $\mathfrak{G}$ a *direct graph*

Example 0.3.2: Graphs

put different types of graphs as example? infinite, connected, cyclic, tree, completed

1. here + visual
2. here + visual
3. here + visual
4. here + visual

(define endpoints of a vertex, adjacent vertices, and if $e = \{u, v\}$ e is also called the incident of u and v)

(define simple graph: E has no repeated elements. If E can have repeated elements, it's a multigraph)

Definition 0.3.3: Graph Isomorphism

Let $\mathfrak{G} = (V_{\mathfrak{G}}, E_{\mathfrak{G}})$ and $\mathfrak{H} = (V_{\mathfrak{H}}, E_{\mathfrak{H}})$ be two graphs. A function $f : V_{\mathfrak{G}} \rightarrow V_{\mathfrak{H}}$ is called a *graph isomorphism* if for every vertex adjacent vertices $v_i, v_j \in V_{\mathfrak{G}}$, we have

$$(v_i, v_j) \in E_{\mathfrak{G}} \Leftrightarrow (f(v_i), f(v_j)) \in E_{\mathfrak{H}}$$

that is, f preserves adjacencies between vertices.

If $\mathfrak{G}$ and $\mathfrak{H}$ are isomorphic, we write $\mathfrak{G} \cong \mathfrak{H}$, and we say that $\mathfrak{G}$ is isomorphic to $\mathfrak{H}$

Example 0.3.4: examples of Graph Isomorphisms

here

These will come back in a exercise ref:HERE

0.3.1 Lattices

(These I think are important because they will be periodically mentioned. From the beginning with the partial ordering of subsets via inclusion, to using Zorn's lemma, to group/ring/module lattices, to the correspondence between the lattice of divisibility in UFDs and multi-sets of irreducible elements, to the many galois correspondences we shall see, to Quivers)

Grillet has a good section on Lattices, so does Lang.

Exercise 0.3.1

1. tree is a-cyclic
2. probably counting vertices and edges problems?
3. other simple and fun graph theory problems?

0.4 Categories

In the last couple of section, we have been focusing on understanding mathematical objects, in particular, we tried constructing different types of objects (like sets and functions, numbers, graphs, and sets with binary relations) and tried showing some properties of these objects. In this section, we will focus less on the mathematical object, and more on the *relation* between mathematical objects that preserve some of the same “structure”. The nouns “structure” and “relation” will be given more precise meaning through the use of *categories*.

Categories have a reputation to be difficult to understand – a reputation I (the author of this book) feel comes from an overly abstract introduction at too early a stage in the pedagogical journey. The hardest part about categories is a shift in perspective in how to think about mathematical objects, and having the sample space to intuit why this shift brings great value. Thus, before getting into the theory of category theory, let’s first give an analogy to get a more intuitive understanding in what we are trying to achieve:

Let’s take any Language (ex. English). We can study any given word to get some properties of it (for example, its grammatical properties, syntactic correctness, its phonetic properties in a dialect, and so forth). These give information about words, however what we usually care about is not the properties of a word, but how words *relate* to other words. The meaning behind a sentence doesn’t come just from the individual words, but from how the words “come together”: This is called semantics²². As an example of semantics, we can even see that the meaning of words is defined in relation to other words in a sentence; for example the word *cold* usually means the absence of heat, but in the sentence “those people are giving him a cold stare”, I am not referencing to the temperature of their gaze, but some lack of friendliness in their feelings towards him.

If we look at language through the lens of the relation between words, we can get some insights on how language is used. An excellent example of this is translation. Let’s say we’re trying to translate from English to Sanskrit (or really any two languages). Both English and Sanskrit are languages, and so they have words (i.e. the objects) and rules (i.e. the relation between objects). Don’t worry for a moment about exceptions to rule in particular languages (If it helps the reader, pretend they don’t exist for the sake of the analogy), the main idea is that English and Sanskrit (and all languages) have words and rules that govern their use. When we compare English to Sanskrit, what we will do is see how the relations of the words in English (ex. conjugations, articles, etc.) link to those in Sanskrit (ex. declension, compounds, etc.). In a sense, we don’t care about what the particular words of the respective language are. What we care is to find the word form one language that will create the same relations in the other language.

For example, If I say “dog” in English, then the sentences “The dog barked”, “I took the dog for a walk”, or “Many dogs like to chase cats” all make sense, while “I dog the took for a walk” is not a sentence that makes sense, since the position of the word “dog” and “take” are wrong given the grammatical rules in the English languages. In Sanskrit, the grammatical rules are different: If we directly translate the sentence “I dog the took for a walk” to Sanskrit, the sentence will actually make sense! That is because the rules of Sanskrit are different than English. In fact, it is a little more subtle than this: There is no “a” (like “a walk”) or “the” in Sanskrit – if you’re translating from English to

²²Semantics is the study of the meaning that sentences (or any discourse) carries

Sanskrit, we can't replace "a" with a Sanskrit word (since non exist). Instead, we will make sure that the words we choose in Sanskrit when translating the English sentences will have the appropriate *relations* in order to capture the same meaning

The entire point of this analogy is to show that it is useful to understand objects not how they "intrinsically" are, but how they "in relation" to other objects ²³. The analogy showed that an advantage of this approach comes from a better understanding of how to translate sentences. Similarly, thinking about mathematical objects by how they relate to one another will let us be able to compare object by seeing if one mathematical object has the same type of relations with other objects.

In fact, we have already done this before: when we worked with ordered structures (ex. $(\mathbb{N}, \leq)$), we studied objects by how they related to other objects. We can even make definitions based on how objects relate. For example, in $(\mathbb{N}, \leq)$, define the *smallest* to be the object x such that for every other element $y \in \mathbb{N}$, $x \leq y$. Then it is clear that 0 is the smallest object. However, in $(\mathbb{Z}, \leq)$ no object is the smallest object, because for any $x \in \mathbb{Z}$, there exists $x - 1 \in \mathbb{Z}$ such that $x - 1 \leq x$. Another example of a definition based on how an object relate, let X be the set for which any function from X is *always* injective. Then X is any singleton (exercise ref:HERE).

In the following section, we will try to more rigorously define a way to talk about the relation between objects

0.4.1 Important Concepts

This section is very much a cursory glance of the elementary concepts of category theory for the sake of exposure rather than for internalizing the definitions and working through problems. Let's start by giving a high-level idea of that a category is. This definition should hold both information about objects and the relation between objects:

Definition 0.4.1: Pre-Definition Of Category

A Category $\mathbf{C}$ is a collection^a of objects and relations between objects:

1. We denote the collection of objects of $\mathbf{C}$ by $\text{Obj}(\mathbf{C})$.
2. For any two objects $A, B \in \text{Obj}(\mathbf{C})$, $\text{Hom}_{\mathbf{C}}(A, B)$ denotes the collection of relation between objects, which we call *morphisms*. In particular, if $f \in \text{Hom}_{\mathbf{C}}(A, B)$, then f is called a morphism. Morphisms are *directional*, meaning that if $f \in \text{Hom}_{\mathbf{C}}(A, B)$, it is *not* the case that $f \in \text{Hom}_{\mathbf{C}}(B, A)$ (unless $A = B$)
If $f \in \text{Hom}_{\mathbf{C}}(A, B)$, we shall write $f : A \rightarrow B$ to emphasize the directionality.
3. If $f \in \text{Hom}_{\mathbf{C}}(A, B)$, and $g \in \text{Hom}_{\mathbf{C}}(B, C)$, then there must exist a morphism $h \in \text{Hom}_{\mathbf{C}}(A, C)$ that we associate g and f ; we'll write $h = gf$, $h = g \cdot f$, or $h = g \circ f$, and call h the *composite morphism* of f and g .

^athe word *collection* is used instead of *set* for a technical reason. More on that after this definition

The fact that we said *collection* instead of *set* is because of a quirk of logic. In particular, once we properly define what a category is, we will show that sets and functions between sets will form a

²³The philosophy behind this idea is called *structuralism*

category. Then $\text{Obj}(\mathbf{Set})$ will have to be a *set of all sets*, but it can be proven that no such set exists! This is called *Russel's Paradox*, and is addressed in chapter ???. The morphisms in $\text{Hom}_{\mathbf{C}}(A, B)$ cannot be chosen arbitrary, they shall have to satisfy the following properties:

1. (identity morphism) For each $A \in \text{Obj}(\mathbf{C})$, there must at least exists morph $1_A \in \text{Hom}_{\mathbf{C}}(A, A)$ such that for any $f \in \text{Hom}_{\mathbf{C}}(A, X)$ and $g \in \text{Hom}_{\mathbf{C}}(X, A)$ (for any $X \in \text{Obj}(\mathbf{C})$), we have:

$$f \text{id}_X = f \quad \text{and} \quad \text{id}_X g = g$$

2. (associativity) For $f \in \text{Hom}_{\mathbf{C}}(A, B)$, $g \in \text{Hom}_{\mathbf{C}}(B, C)$ and $h \in \text{Hom}_{\mathbf{C}}(C, D)$, then if $\alpha = gf$ and $\beta = hg$, then:

$$h\alpha = \beta f \quad \text{i.e.} \quad h(gf) = (hg)f$$

The motivation behind these rules will come in chapter ??.

Example 0.4.2: Category of Set

The prototypical examples of a category is $\mathbf{Set}$ with object as sets and morphisms as functions, namely for any two sets $X, Y \in \text{Obj}(\mathbf{Set})$, $\text{Hom}_{\mathbf{Set}}(X, Y)$ is the collection of all functions from X to Y . For any $X \in \text{Obj}(\mathbf{Set})$, the identity function is the identity morphism, and section 0.1.5 showed functions are associative.

We shall cover more examples when there is more material to cover in chapter ?? and more advanced examples in part ??.

Let us now go over some important concepts that shall come up continuously throughout the entire textbook.

Definition 0.4.3: Isomorphism

Let $\mathbf{C}$ be a category (the reader can keep in mind $\mathbf{Set}$). Then we shall say that $X, Y \in \text{Obj}(\mathbf{C})$ are *isomorphic* if there exists morphisms $f \in \text{Hom}_{\mathbf{C}}(X, Y)$ and $g \in \text{Hom}_{\mathbf{C}}(Y, X)$ such that $fg = \text{id}_X$ and $gf = \text{id}_Y$. We shall sometimes abuse notation and simply write $fg = gf = \text{id}$. If X, Y are isomorphic, we shall write:

$$X \cong_{\mathbf{C}} Y$$

Example 0.4.4: Isomorphisms in Set

Let $\mathbf{C} = \mathbf{Set}$. Show that X, Y are isomorphic if and only if $|X| = |Y|$. This shows that the relations in the category $\mathbf{Set}$ “capture” or “remember” cardinality information of sets. If we had different morphisms, we can have different types of information.

Recall that what we care about in category theory are the relations between objects and not necessarily the objects themselves. As a consequence, results proven in $\mathbf{Set}$ shall be true for all sets of the same cardinality. More generally, we shall see that results proven in category for an object X shall be true *up to isomorphism*, that is if they are true for X , they are true for Y such that $X \cong_{\mathbf{C}} Y$.

The reader may think to themselves that cardinality information is not much to preserve, for example we may have two sets of the same cardinality that satisfy different distinct predicates. This is because $\mathbf{Set}$ is a very simple category and each morphism holds little information^a. We

shall often extend the objects and morphism of **Set** to hold more information.

^aThe knowledgeable category theorist may complain that by Yoneda's lemma, any locally small category can embed into **Set** and so the category is far from being simple. The aforementioned simplicity is from the basic description and isomorphisms classes rather than the complicated subcategories we may produce

The next important idea is that of *commutative diagram*. These shall represent how we can relate objects:

Definition 0.4.5: Commutative Diagram

Let **C** be a category. Then a *commutative diagram* is a graph where each node is an object $X \in \text{Obj}(\mathbf{C})$ and each edge is a morphism (with appropriate domain and codomain) where for each closed loop in the graph (disregarding orientation), the composite morphisms must agree.

To elucidate the last condition, consider:

$$\begin{array}{ccc} & Y & \\ f \nearrow & & \searrow g \\ X & \xrightarrow{h} & Z \end{array}$$

This forms a closed path, and hence composition must agree, and so $gf = h$. Another common example is:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow h & & \downarrow g \\ Z & \xrightarrow{i} & W \end{array}$$

Then:

$$gf = ih$$

As we care about the relations between objects, commutative diagrams will be the building for most categorical theorems. A lot of results shall come down to saying “the following diagram commutes”. Let us apply this to the first concept of category theory: *universal properties*. A universal property shall often be a property that captures the *essence* of a predicate in such a way that we can ask whether an object in different categories have or don't have the property. This is part of where the word *universal* comes from; these shall represent properties that can be described *universally between categories*. The full generality of universal properties is something we shall delay for chapter ???. For now, let us give an important example of the *universal property of quotients*. We shall show what they exist in **Set**, and in each part of the book we shall see how the majority of categories we work with shall have objects satisfying this universal property.

The name suggests some notion of “dividing”. This is certainly strongly related to dividing in the traditional numerical sense, however the connection to dividing would be too large of a build up for the expository nature of this section, and so it will have to be relegated to chapter ??. Instead, the reader should think of *equivalence relations* of a way of dividing up a set. The universal property of quotients in **Set** shall show that there is an object that represents the equivalence relation:

Theorem 0.4.6: Universal Property Of Quotient Sets

Let A be a set and $\sim$ be an equivalence relation on A . Let $f : A \rightarrow Z$ be any set function such that $a \sim b \Rightarrow f(a) = f(b)$. Then there exists a unique map φ from $A/\sim$ to Z such that the following diagram commutes

$$\begin{array}{ccc} A/\sim & \xrightarrow{F} & Z \\ \pi \uparrow & \nearrow f & \\ A & & \end{array}$$

Proof :

Let's say $f : A \rightarrow Z$ respected where $a \sim b \Rightarrow f(a) = f(b)$. we want to ask if there exists a unique map such that

$$f = F \circ \pi$$

that is

$$f(x) = F(\pi(x))$$

Consider the map $F : (A/\sim) \rightarrow Z$, $F([x]) = f(x)$. This map is well-defined since if $[x] = [y]$, then $x \sim y$ so $f(x) = f(y)$.

Furthermore, this map is unique. If $F, F' : (A/\sim) \rightarrow Z$ were two well-defined maps, then it must be that $F(\pi(x)) = f(x) = F'(\pi(x))$ making $F = F'$, showing uniqueness, completing the proof.

Essentially, given an equivalence relation $\sim$ on a set A it can be “represented” by the object $A/\sim \in \text{Obj}(\mathbf{Set})$ which captures the right amount of information such that any other object that tries to capture it is always linked to $A/\sim$. In other words, $A/\sim$ is in a sense always in an object that tries to represent the equivalence classes of A .

This will conclude our quick exposition of category theory. In chapter ??, we shall properly introduce the theory of category theory at the “elementary level”. In part ??, we shall properly cover category theory in its full generality and introduce the important major results that students ought to carry throughout their mathematical career.

0.5 Matrices

This section can be comfortably skipped until chapter ?? . It will also be referenced in section ?? for examples of groups, but it is skippable without any loss (since it's just one out of over a dozen example of groups that will be presented).

In this book, we will work with many structures. Some of them will be very different from one another, while others share striking similarities. Many of these more similar mathematical structures will be able to be “represented” by matrices. We will define more rigorously for something to be “representable” by matrices later – for now we want to gain some basic appreciation and understanding of how to use matrices.

(History going back to China, this is the early motivator for matrices compared to the “new” one of being a useful classification theorem. Define a matrix. Show it's used to solve systems of linear equations. Explain Gaussian elimination)

(Later, Gauss developed a way to multiply matrices. The intuition shall come in chapter ??)

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